SMART INDIA HACKATHON 2026

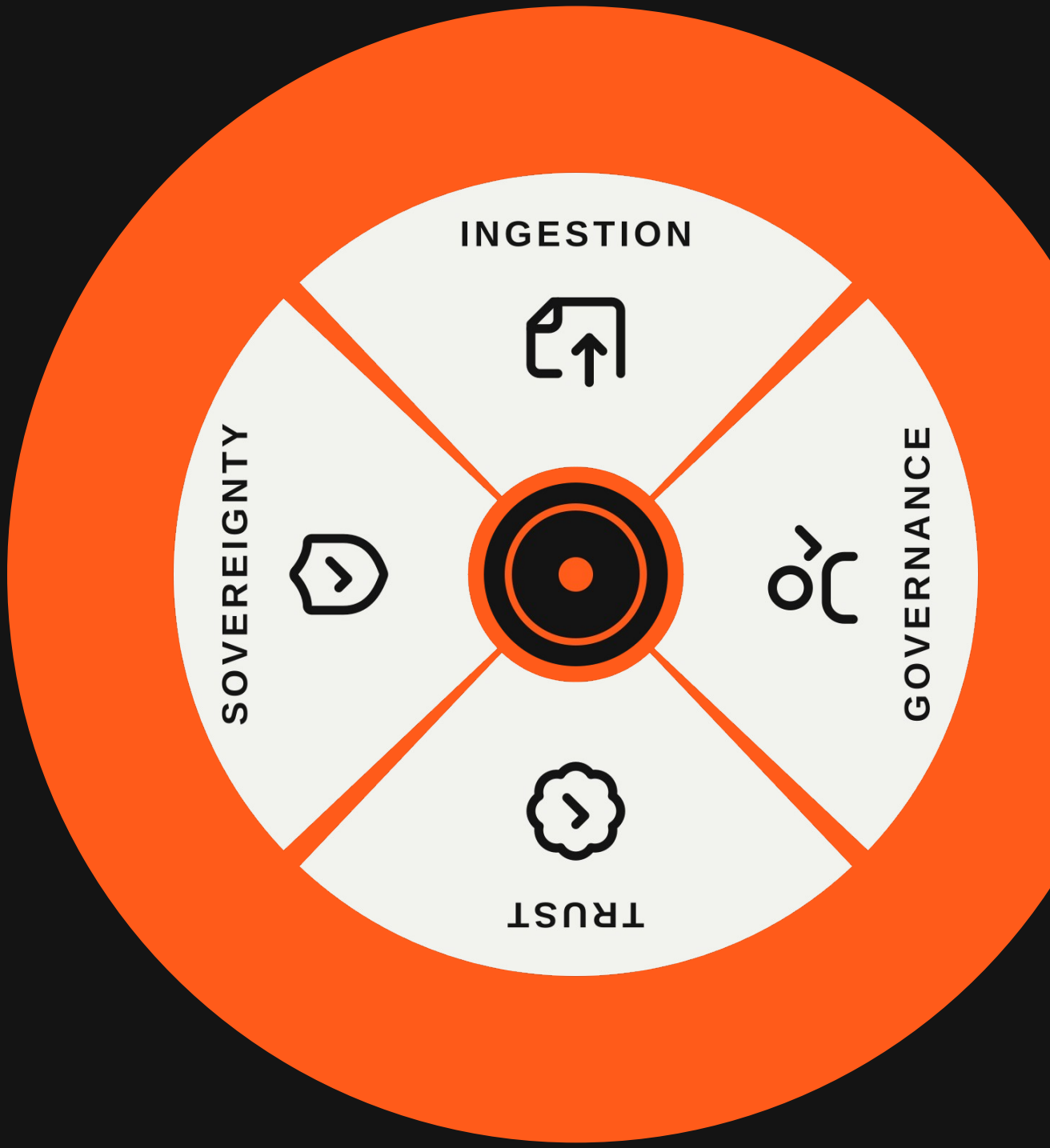
SOVEREIGN

AGENTIC AI WORKBENCH

Sovereign On-Premise Agentic AI Workbench using Open-Weight Multimodal LLMs for Confidential Industrial Work

PS ID	THEME	CATEGORY
SIH26117	Smart Automation	Software

TEAM KERN / 4CE (ACE)



OUR SOLUTION



SOVEREIGNTY

Local LLMs → data stays inside

INGESTION

Legacy files → usable intelligence

GOVERNANCE

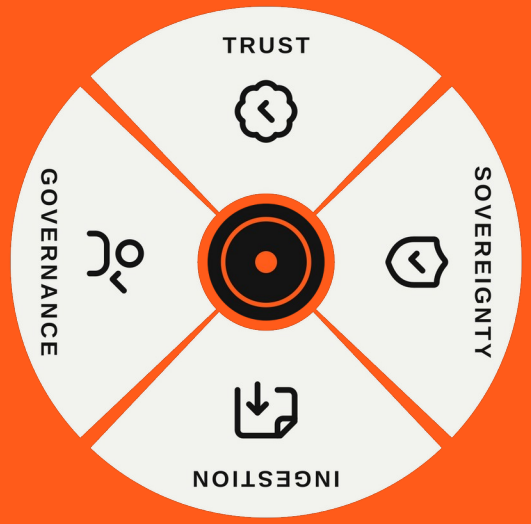
Tool execution → human approval

TRUST

Generated answer → source-verified

WHY DIFFERENT

Three design choices make agentic AI safe to run on confidential data.



Isolated sandbox execution

AI-generated code → Isolated environment → **Controlled execution**

Centralised AI control

Every action → Control layer → Authorization → **Execution**

Sensitive actions → mandatory human approval

Model-independent architecture

Qwen · Gemma · Mistral · Other open-source models

Change the model, not the architecture.

TECHNICAL APPROACH

Language & runtime
Python · Bash

Application platform
FastAPI · Uvicorn · Socket.io · SvelteKit · JWT session auth

AI / ML
Bionic · Qwen models · Gemma models · nomic-embed-text-v1.5

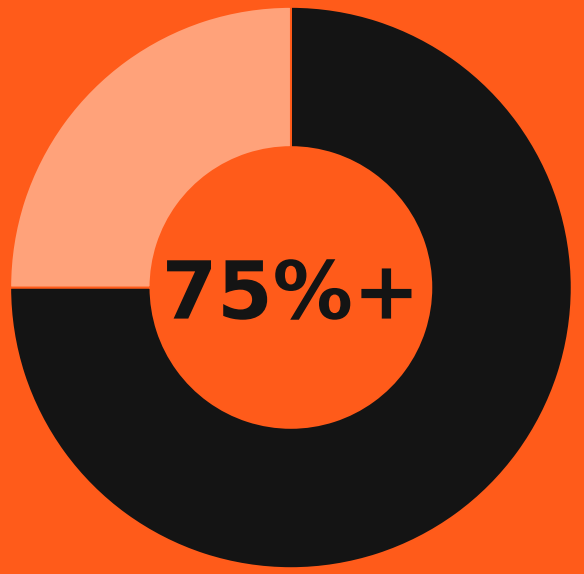
Data & storage
SQLite · ChromaDB

Execution & isolation
Docker · hardening flags

Document generation
python-docx

Deployment & tooling
Custom test harness

BUILD STATUS



of the prototype is already built

View the GitHub repo



FEASIBILITY & VIABILITY



RISK	MITIGATION
Cloud dependency Data could leave the enterprise	**Local-only stack + outbound paths disabled**
65K context window KV cache exceeded 6 GB VRAM → CPU fallback → 20+ min inference	**8K context fits VRAM → GPU inference restored**
AI hallucination Generated findings could be incorrect	**ULTRON verification + bounded replan**
Generated-code risk AI code could reach the host or network	**Hardened Docker sandbox → isolated execution**

RESOURCES & ECONOMICS



RESOURCE REQUIREMENTS

- **Local compute + GPU (recommended)**
- **Docker + local model server**
- **Persistent storage + enterprise data**
- **Proven frameworks**

ECONOMIC APPROACH

Lower recurring AI costs

Potentially lower costs for sustained, high-volume workloads

Enterprise licensing

Monetised through licensing plus setup and integration

Recurring revenue

Support, maintenance and custom agents

IMPACT & BENEFITS

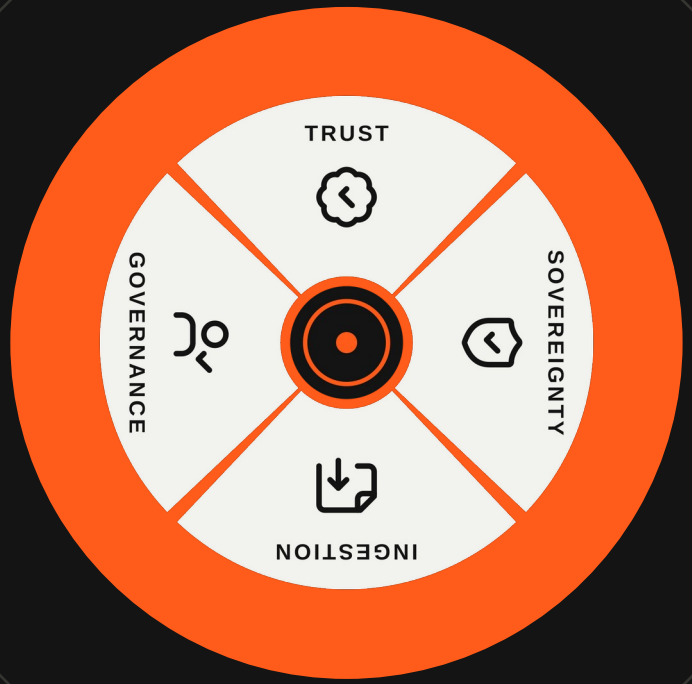
Real value, higher control, a safer tomorrow.

01 CONVERSATIONAL IMPACT

- Safer AI interactions
- Human oversight

02 FASTER RESEARCH

- Private AI research
- Controlled experimentation



03 NATIONAL IMPACT

- AI & data sovereignty
- Reduced external dependence

04 TECHNICAL IMPACT

- Local LLM + RAG
- Multi-agent orchestration
- Sandbox + verification · auditability

05 ECONOMIC & INDUSTRIAL IMPACT

Lower recurring AI costs · secure enterprise adoption · regulated industries

FUTURE PROSPECTS

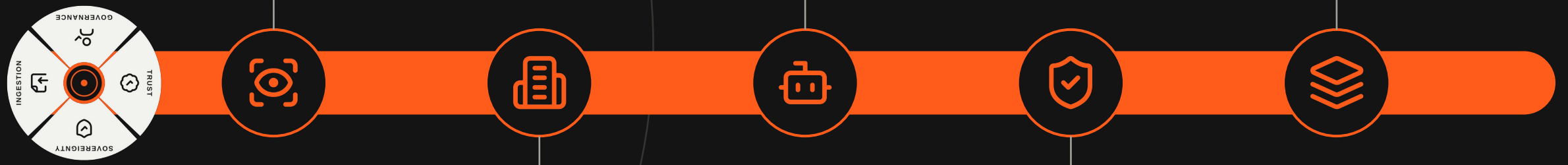
Multimodal enterprise AI
Vision, audio, video & complex-document understanding

Multi-site deployment
Local AI at every site with centralised governance

Industry-specific agents
Manufacturing, healthcare, finance, government & defence

Continuous security & governance
Policy enforcement, monitoring, audit & compliance

Enterprise AI control layer
Evolve into the secure orchestration and governance layer for enterprise AI



RESEARCH & REFERENCES



SUPPORTING RESEARCH PAPERS

Attention Is All You Need

Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks

Self-Refine: Iterative Refinement with Self-Feedback

AutoGen: Next-Gen LLM Applications via Multi-Agent Conversation

TECHNICAL DOCUMENTATION

Sovereign AI Project Deep Research

Master Architecture + Phase Specifications

Verification Reports + Security Runbooks

RUNNING.md + Sovereign Configuration

MARKET RESEARCH

Inference Economics of Enterprise Coding Agents — 2026

Hardware Economics: Cloud vs On-Prem TCO — 2026

On-Premise vs Cloud: Generative AI TCO — 2026

LINKS

GitHub repo

Demo video

PPT video explanation

YOUR DATA STAYS YOURS

Sovereign. Secure. Verifiable. Actionable.

- Turns private organisational knowledge into actionable intelligence.
- Automates real workflows while keeping humans in control.

Powerful enough for enterprises. Controlled enough to trust.

THANK YOU Team Kern • 4CE (ACE)

